

SYLLABUS :-

Soil Genesis and Classification: Weathering, formation and composition of soil, classification of soils. **Soil Colloids:** Inorganic and organic colloids, their nature and physico-chemical properties, ion exchange phenomena. **Mineral Nutrition of Plants:** Essential elements, chemical fertilizers, organic manures. **Movement of chemicals in soil-root continuum** **Physics of Soil:** Texture, structure and basic physical properties, retention and movement of soil water, solute transport, gaseous exchange, soil strength and thermal properties. **Mechanics of Soil:** Dynamic properties of soil, stress-strain relations, yield and failure, soil compaction and consolidation, earth pressure and stability of slope. **Soil Tillage:** Types of tillage, influence of tillage on soil properties and crop performance. **Soil Management:** Management of acid, saline and sodic soils.